

Purna Ananthkrishnan

Indian Institute of Technology Madras (India)

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Education

Indian Institute of Technology, Madras

Dual Degree (B.Tech & M.Tech) with Honours in Aerospace Engineering

2022 - Present

CGPA: 9.81/10

Kendriya Vidyalaya IIT Powai

All India Senior School Certificate Examination

2020 - 2022

Grade: 98.6%

Kendriya Vidyalaya IIT Powai

All India Secondary School Examination

2010 - 2020

Grade: 97.2%

Scholastic Achievements

- Currently **ranked 1st** among students in the Dual Degree program in the Aerospace Engineering department.
- Awarded the **Sri K Krishnamurthi Prize** for securing the second highest CGPA at the **65th Institute Day**
- Secured a **perfect 10/10 Grade Point Average** in Semesters II & III, reflecting consistent academic excellence.

Conference Publications and Presentations

- **Purna Ananthkrishnan**, Siva Vayala and Sriram Rengarajan. "Interaction of Moving Expansion Waves: A One-Dimensional Method of Characteristics Framework." *26th International Shock Interaction Symposium, South Korea**
 - Dwarakesh Madavan, **Purna Ananthkrishnan**, Samiayyan Pandian, Malsur Dharavath and Sriram Rengarajan. "Experimental Investigation of Shock-Cavity Shear Layer Interaction in a Confined Supersonic Flow over a Tapered Cavity with Gas Injection." *Proceedings of the 35th International Symposium on Shock Waves, Brisbane, Australia***
- (* :Accepted for Oral Presentation, ** :Revisions Submitted, Under Review)

Research Experience

Unsteadiness in Crossing Shockwave Boundary Layer Interaction

Master's Thesis, Guide: Prof. R. Sriram

May 2026 – Present

Gas Dynamics Laboratory, IIT Madras

- Investigating the governing mechanisms of unsteadiness in the 3-D crossing shock wave boundary layer interactions.
- Exploring the possibility of hysteresis being associated with the transition from regular reflection to Mach reflection.
- Employing Schlieren and oil-flow visualization & unsteady pressure measurements, in a supersonic wind tunnel facility.
- Implemented Background-Oriented Schlieren (BOS) for an axisymmetric case and are extending it to tomographic-BOS.

Interaction of Moving Expansion Waves via Method of Characteristics

Undergraduate Research, Guide: Prof. R. Sriram

Aug 2025 – Apr 2026

Gas Dynamics Laboratory, IIT Madras

- Applied the method of characteristics to develop a 1-D theoretical framework to solve expansion wave interactions.
- The developed solver produced physically reasonable results for head-on interactions of two moving expansion fans.
- Currently validating the results against an in-house compressible flow solver, to establish the framework's accuracy.

Reduced Order Modelling of Acoustic Response of a Combustor

Research Internship, Guide: Prof. Kurose Ryoichi

May 2025 – Jul 2025

Thermal Physics Laboratory, Kyoto University

- Selected for Kyoto University's Short Term Academic Research Program; presented a poster on the results obtained.
- Investigated the acoustic response and eigenfrequencies of a simplified combustor model using COMSOL Multiphysics.
- Developed a reduced-order model via a Green's function approach to determine pressure distributions using eigenmodes.
- Successfully validated the developed model against simulation data, predicting pressure distribution at any frequency.

Shock-Cavity Shear Layer Interaction with Gas Injection

Undergraduate Research, Guide: Prof. R. Sriram

Nov 2024 – Apr 2025

Gas Dynamics Laboratory, IIT Madras

- Conducted experiments to study shock-cavity shear layer interaction over a tapered cavity, with passive air-jet injection.
- Performed proper orthogonal decomposition of Schlieren images and power spectral density analysis of pressure data.
- Observed an absence of Rossiter modes, indicating suppression of the acoustic feedback loop and resonant behaviour.

Modelling of the Lifted Temperature Minimum Phenomenon

Research Internship, Guide: Prof. K.R. Sreenivas

May 2024 – Jul 2024

Fluid Mechanics Laboratory, JNCASR

- Awarded the summer research fellowship by Jawaharlal Nehru Institute for Advanced Scientific Research (JNCASR)
- Modelled nocturnal boundary layer temperature profiles with aerosol heat transfer, demonstrating a lifted temperature minimum

Relevant Coursework, Skills, and Teaching Experience

Core Courses: Aerodynamics, Gas Dynamics, Propulsion, Flight Dynamics, Design and Fabrication of UAV/MAVs

Aerodynamics Electives: Boundary Layer Theory, Shockwave Dynamics, Waves in Fluids, Incompressible Turbulence

Computational Fluid Dynamics Electives: Fundamentals of CFD, Computational Aerodynamics, Advanced CFD

Other Electives: Nonlinear Dynamics, Data-Driven Methods, Quantum Computation & Information, Active Fluids

Programming: C/C++, Python, MATLAB, FORTRAN90

Software: Ansys Fluent, COMSOL Multiphysics, Fusion360, XFOIL, XFLR5

Experimental Techniques: Schlieren and Shadowgraph Imaging, Oil-Flow Visualisation, Background-Oriented Schlieren

Teaching Assistantship: AS2100: Basic Aerospace Engineering Laboratory (Jul – Nov 2026)

Projects

High-Order 3-D Multiblock Compressible Flow Solver in Fortran

Course Project, Advanced Computational Fluid Dynamics

Jan 2026 – May 2026

- Developed a 3-D compressible flow solver in Fortran90, with high-order compact & explicit finite-difference schemes
- Validated the DNS solver against canonical benchmark cases – Taylor-Green Vortex and inviscid vortex convection
- Incorporated RANS with $k - \omega$ and the **novel** $k - \omega_0$ turbulence model to simulate the viscous flow over a flat-plate.
- Demonstrated significant improvement in numerical stability, better accuracy on coarser grids for the $k - \omega_0$ model.

Explicit Euler Solver for Compressible Flow Past a Bump

Course Project, Computational Aerodynamics

Jul 2025 – Nov 2025

- Developed an explicit finite-volume incompressible Euler solver using the artificial compressibility method in MATLAB
- Extended the solver to compressible flow, implementing Roe, Van Leer, and AUSM schemes for interface flux evaluation
- Simulated the flow over a bump and verified the accuracy of the solver across a range of Mach numbers from 0.4-1.4

Fixed-Wing UAV for Long-Range Maritime Surveillance

Course Project, Design and Fabrication of UAV/MAVs

Jul 2025 – Nov 2025

- Designed and fabricated a fixed-wing UAV with a 2 m wingspan and 8 kg MTOW for long-range maritime surveillance
- Performed a thorough aerodynamic, structural, and stability analysis to validate the design for the mission requirements.
- Fabricated the UAV airframe with all structural elements, primarily using aluminium sheets and 3-D printed components

Coupling of Structure and Wake Oscillators in Vortex-Induced Vibrations

Course Project, Nonlinear Dynamics

Jan 2025 – May 2025

- Modelled the coupled structure-wake oscillator system from Facchinetti et al.(2004) to study vortex-induced vibrations.
- Validated the model's predictions of hysteretic behaviour and extended the model to incorporate nonlinear coupling.

Parachute Mechanism to Enhance Drone Safety

Internship, AeroDome Technologies Pvt. Ltd.

Nov 2023 – Jan 2024

- Conceptualized & designed a mechanical parachute deployment system, triggered by IMU & barometric sensor inputs.
- Implemented an Arduino-based control code to ensure reliable, timely deployment under uncontrolled flight conditions.

Fixed-Wing UAV for Medical Supply Delivery

Project under the Center for Innovation, IIT Madras

Jun 2023 – Mar 2024

- Designed and developed a prototype fixed-wing long-range UAV for delivering medical supplies to inaccessible areas.
- Performed weight estimation and constraint analysis using Python codes; designed & analysed the wing using XFLR5

Extracurricular Activities

Inter IIT Tech Meet 14.0: Represented IIT Madras's Tech Contingent at IIT Patna, working on a short take-off aircraft design problem; the solution was obtained by including CFD-based aerodynamic analysis and design optimisation.

Avanti Fellows (NGO), IIT Madras: Headed a team with 40+ members, guiding underprivileged students to prepare for college entrance examinations, with the mentees securing seats at top engineering universities across India.

Classical Dance: Trained in Bharatanatyam for 10+ years and performed at multiple events on various occasions.